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CHAPTER 1

Introduction

The questions on how did the universe came to be, why is the universe as such, and what is the universe heading towards falls upon cosmology to answer. Cosmology is the study of the universe on large scales in an attempt to answer the latter questions cite maybe Peebles. This pursuit started as a data-starved research cite and transformed into a data-rich endevour as a result of a platitutde of imaging surveys such a-s the LSST Vera Rubin observatory, Euclid, and Nancy Roman cite. The qualitative questions, with which this paragraph started,< transcribe in the statistical sense to be what is the cosmological model that best describes the universe on throughout length scales and energy scales.

The first attempt to describe the universe as a whole came through the lens of the general theory of relativity where Robertson and Walker cite built on the work of Friedmann and Lemaître to describe geometry of the universe through what is now known as the Friedmann-Lemaître-Robertson-Walker (FLRW) metric. We discuss this metric further in 2.2. This metric was largely motivated by the observations of Edwin Hubble who saw that galaxy further away from the Earth are moving faster than galaxies closer to Earth cite. Figure shows the plot from Hubble's work in 1929.

Further observations of the magnitude of distant galaxies have shown that the universe is expanding cite. The general theory of relativity attributes this expansion to the energy density content of the universe. Therefore, the question of what is driving the expansion of the universe and what are the componenets that contribute to the energy budget of the universe represent a crucial question in cosmology for which multiple cosmological models have been suggested (Di Valentino et al., 2025). In fact, the cosmological model has been motivated by both cosmological and astrophysical observation. Some of these observation are the measurement of galaxy rotation curves by Vera Rubin cite that hinted at an additional matter component that does not interact with light and the discovery of the cosmic microwave background (CMB) radiation that hinted at hot and homogeneous early universe.cite. Furthermore,early observaiton of clusters of galaxies have shown that the spatial distribution of galaxy deviates from randomness. This was shown by Peebles cite. By calculating the two point correlation function, there was clearly a signal in the way that galaxies are distributed which points out to an underlying mechanism in the formation of these galaxies (Totsuji and Kihara, 1969). The development of obsrevational techniques and instrumentation allowed the launch of the Center for Astrophysics Redshift Survey (Davis, Huchra et al., 1982) which showed that galaxy positions are correlated on large scales and that there is a structure in the universe (Davis and Peebles, 1983).

In the early 1990s, cosmology experienced a paradigm shift with the launch of the Cosmic

Background Explorer (COBE) sattelite (Boggess et al., 1992). The COBE mission provided the first detection of primordial CMB anisotropies and further confirmed that the universe is flat (Smoot, 1999). In the context of the FLRW matric, the flatness of the universe requires a total sum of the relative density parameters to be 1 (within error bounds). Therefore, in addition to the matter component (dark and baryons), there should be a third component that can account for the missing energy density. Using supernova data from the Hubble Space Telescope (Bahcall and Spitzer, 1982), A. G. Riess et al. (1998) found a strong evidence for an expanding universe universe with negative acceleration and supported the findings of the COBE mission for the existance of additional density component different from matter which is driving the expansion of the universe. Hereafter, the term 'dark energy was coined to refer to this component (Turner, 2001). Wiht this, the ΛCDM model was born, and the efforts to obtain a precise estimates of the cosmological parameters intensified (Bennett et al., 2003; Pope, 2004).

While early experiements relied on CMB, galaxy number counts, or supernova to infer cosmological parameters, gravitational lensing was in development as a tool to constrain the cosmological parameters. Gravitational lensing is the distortions of light geodesics by gravity when light travels from a distant source in the universe towards an observer on the Earth (Schneider, Ehlers and Falco, n.d.). On the level, of a galaxy, gravitional lensing effects manifest themselves as a distortion of the shape of the galaxy and a change in the angular position with respect to a particular observer. Kaiser and Squires (1993) showed that these distortions could be used to obtain a map of the integrated mass density along the line of sight. The use of galaxy shape distortions for cosmology was a direct result of the Kairser-Squire inversion formula where Seitz and Schneider (1996) realised that the expectation value of galaxy shape is an unbiased estimate of the shear in the weak field limit. In principle, it is possible to relate the power spectrum of galaxy shapes with the matter power spectrum, offering a concrete method of probing cosmology by studying the correlations between galaxy shapes (Bartelmann and Schneider, 2001). This new probe of the large scale structure is called cosmic shear and was used by Maoli et al. (2001) to obtain constraints on the cosmological parameters. These constraints were not as tight as the ones obtained by COBE, however, it paved the road towards using cosmic shear to constrain the cosmological model. The advantage of cosmic shear as an integrated quantitity along the line of sight mean that it is sensitive to structure growth in the late universe. Therefore, it represent a very good tool to constrain the equation of state for dark energy (Hu and Tegmark, 1999; Jain and Taylor, 2003).

After that, multiple galaxy surveys used cosmic shear for cosmilogical parameter inference, and numerous statistical tools were developped to deal with the increasing data size (Schneider, Van Waerbeke et al., 1998; Schneider, Eifler and Krause, 2010). Likewise, with increasing data size, the noise level in a cosmological signal increases and the need to mititgate the impact of the systematic and measreument uncertainties is imperative. In this work, we focus on tackling on of the most crucial systematics, namely baryonic feedback on the large scale structure.

Baryonic feedback are the set of baryonic physics processes that impact the distribution of the matter (Rudd, Zentner and Kravtsov, 2008). These processes include, but not limited to, supernova feedback, activate galactic nuclei feedback, and star formation. By analying hydrodynamical simulations, Daalen et al. (2011) realised that baryonic feedback affect the matter distribution in the universe at different scales, meaning that their impact should be modelled and integrated . However, the theoretical formalism of structure growth, and therefore the matter power spectrum assumes that structure growth is only driven by gravitional effects, ignoring baryonic feedback (Dodelson and Schmidt, 2026). These uncertainties naturally tunel down to any cosmic shear statistics since it is related to the matter power

spectrum. To mitigate the impact of baryonic,

To estimate the angular power psectrum up to small angular scales (ℓ $\tilde{5}000$), the matter power spectrum should be accuretly modeled up to an equivilant small scales (k 10, $/hMp >^{-1}$)

One of trhe

To unleash the full potential of cosmic shear The theoretical formalism of cosmic shear assumes that structure growth is only driven by gravitional effects. However, hydrodynamical simulations has shown that baryonic feedback affect the matter distribution in the universe at different scales Daalen et al. (2011), and ignoring the impact of baryonic physics leads to biased estimate of the cosmological parameters in weak lensing analysis. The problem of baryonic phyiscs is very unique to using cosmic shear as a probe and multiple methods have been suggested to mitigate their impact.

For example, statistical scale cuts (Krause et al., 2021), theoretical error covariance (m), SZ effect,

Speak about the FLRW metric, Hubble flow, energy content, impact of the Hubble flow and initial conditions on structure formation and evolution. Dark matter and traces of dark matter. Weak lensing, baryonic feedback FRBs, simulation based inference.

- Large Scale Structure
- Lensing
- Weak gravitational Lensing
- Precision Cosmology & tensions
- Baryonic feedback
- FRBs
- systematics of FRBs

CHAPTER 2

Theoretical Background

2.1 The general theory of relativity

2.2 The Friedmann-Robertson-Lemaître-Walker metric

2.3 The cosmological model

2.4 Weak gravitational Lensing

2.5 The impact of baryonic feedback on the large-scale structure

2.6 Fast radio bursts

describe how modelling the PDF is a hard thing to do To forward model observations, we convolve the $DM - z$ represented by $P(DM|z)$ with measurement and systematic uncertainties, in addition to other observational effects. A catalog of detected FRBs is expected to have observed dispersion measure $DM^i_{\rm obs}$, galactic coordinates, and redshifts for a subset of and observations z^i. We assume that each detected FRB was emitted at redshift z^i where the difference between identified and unidentified host can be integrated in how $P(z^i)$ is modeled. Assume that we observe N_{FRB} fast radio bursts signal. We express the observations as the probability of detecting an FRB $DM_{\rm obs}$ given a set of estimated redshifts $\{z^i\}$, can be decomposed using the Bayes rule such that

$$P(DM_{\rm obs}|\{z^i\}) \propto P(\{z^i\}|DM_{\rm obs})P(DM_{\rm obs}) \tag{2.1}$$

Furthermore, if we assume that each observation z_i is independent of other observations we may use the Bayes rule to decompose $P(\{z^i\}|DM_{\rm obs})$ such that

$$P(\{z^i\}|DM_{\rm obs}) = \prod_i P(z^i|DM_{\rm obs}) \tag{2.2}$$

Furthermore, the individual terms in 2.2 could be derived using the joint probability distribution of the observed and the dispersion measure contribution from the large scale structure, namely $DM_{\rm obs}$

and DM such that

$$\begin{aligned} P(z^i|DM_{\text{obs}}) &\propto P(DM_{\text{obs}}|z^i)P(z^i) \\ &= P(z^i)\int P(DM_{\text{obs}}, DM|z^i)dDM \\ &= P(z^i)\int P(DM_{\text{obs}}|z^i, DM)P(DM|z^i)dDM \end{aligned} \tag{2.3}$$

The final expression that relates the probability distribution function of an observation to other modelled terms is

$$\begin{aligned} P(DM_{\text{obs}}|\{z^i\}) \propto \\ P(DM_{\text{obs}})\prod_i P(z^i)\int P(DM_{\text{obs}}|z^i, DM)P(DM|z^i)dDM \end{aligned} \tag{2.4}$$

Furthermore, to model each component of the resulting expression, we start with $P(DM_{\text{obs}})$. There are three components that contribute additively to DM_{obs}: the large scale structure contribution DM_{LSS}, the Milkyway contribution DM_{MW}, and the host galaxy contribution DM_{host}, and the observed dispersion measure is given as

$$DM_{\text{obs}} = DM_{\text{LSS}} + DM_{\text{MW}} + DM_{\text{host}} \tag{2.5}$$

For simplicity, we indicate to the contribution from the large-scale structure simply with DM which is the same in Equation 2.4. Consider the joint probability distribution $P(DM, DM_{\text{MW}}, DM_{\text{host}})$, this can be related to $P(DM_{\text{obs}}) = P(DM + DM_{\text{MW}} + DM_{\text{host}})$ via the constrained integral

$$\begin{aligned} P(DM_{\text{obs}}) &= P(DM + DM_{\text{MW}} + DM_{\text{host}}) \\ &= \int_C P(DM, DM_{\text{MW}}, DM_{\text{host}})\,\mathrm{d}DM\,\mathrm{d}DM_{\text{MW}}\,\mathrm{d}DM_{\text{obs}} \\ &= \int_C P(DM)P(DM_{\text{MW}})P(DM_{\text{host}})\,\mathrm{d}DM\,\mathrm{d}DM_{\text{MW}}\,\mathrm{d}DM_{\text{obs}} \end{aligned} \tag{2.6}$$

Where $C = \{DM, DM_{\text{MW}}, DM_{\text{host}}; DM_{\text{obs}} = DM + DM_{\text{MW}} + DM_{\text{host}}\}$ Similarly, I can model $P(DM_{\text{obs}}|z^i, DM)$ as

$$\begin{aligned} P(DM_{\text{obs}}|z^i, DM) = \int_C P(DM|z^i, DM)P(DM_{\text{MW}}|z^i, DM) \\ P(DM_{\text{host}}|z^i, DM)\,\mathrm{d}DM\,\mathrm{d}DM_{\text{MW}}\,\mathrm{d}DM_{\text{obs}} \end{aligned} \tag{2.7}$$

Each of the terms making up the integral could be further reduced

$$P(DM|z^i, DM) \propto P(DM|z^i)P(DM) \tag{2.8}$$

$$P(DM_{\mathrm{MW}}|z^i, DM) = P(DM_{\mathrm{MW}}) \tag{2.9}$$

$$P(DM_{\mathrm{host}}|z^i, DM) \propto P(DM_{\mathrm{host}}|z^i)P(DM|z^i) \tag{2.10}$$

Keeping in mind, that DM_{host} that we use in 2.4 is flat since we do not have any information of the

CHAPTER 3

Methodology

In this section, we explain the methoddology used in this thesis.

3.1 Modelling the electron power spectrum

3.1.1 The baryonification correction model

3.1.2 The halo model

The Core Cosmology Library (CCL)

3.2 Simulating the dispersion measure signal

3.2.1 the Generator for Large Scale Structure (GLASS)

3.2.2 The electron angular power spectra

3.2.3 The cosmological covariance of the dispersion measure signal

3.3 The systematic uncertainties

It is important here to mention how this is modelled on the level of the simulation and the level of covariance.

3.3.1 Redshift uncertainty

3.3.2 Selection effects

3.3.3 Unlocalised hosts

3.3.4 Milkyway contribution

3.3.5 Host galaxy contribution

3.3.6 The lensing of the Dispersion measure signal

3.4 Simulation-based inference

3.4.1 Approximate Baysian inference

3.4.2 Neural Posterior Estimator

3.4.3 The Sequentual Neural Posterio Estimator

The truncated sequential neural posterior estimator

3.4.4 Flow matching posterior estimator

APPENDIX A

Useful information

In the appendix you usually include extra information that should be documented in your thesis, but not interrupt the flow.

The LaTeX WikiBook (**latexwiki**) is a useful source of information on LaTeX.

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You should probably use `\chapter*` for acknowledgements at the beginning of a thesis and `\chapter` for the end.